

Program-4

Integrate Git with Docker for Source Controlled Application Builds

Folder Structure

Program-4

- Dockerfile
- index.js
- package.json
- package-lock.json
- node_modules

Step 1: Create the program folder

```
mkdir Program-4
```

```
cd Program-4
```

Step 2: Create a Dockerfile

```
1rv24mc012_akanksha@akanshasPC:~$ mkdir Program-4
1rv24mc012_akanksha@akanshasPC:~$ cd Program-4
1rv24mc012_akanksha@akanshasPC:~/Program-4$ nano Dockerfile
```

```
FROM node:18 AS build
```

```
WORKDIR /app
```

```
COPY package*.json ./
```

```
RUN npm install
```

```
COPY . .
```

```
FROM node:18-slim
```

```
WORKDIR /app
```

```
COPY --from=build /app /app
```

EXPOSE 3000

CMD ["node", "index.js"]

Step3: Create an index.js file

```
1rv24mc012_akanksha@akanshasPC:~/Program-4$ nano index.js
```

```
const express=require("express");
const app=express();
const PORT=3000;
app.get("/", (req,res) =>{
res.send("Heloo from git +Docker Integration");
});
app.listen(PORT, () =>{
console.log('Server running at http://localhost:${PORT}');
});
```

Step 4: Create package.json file by running the following command

npm init -y

```
1rv24mc012_akanksha@akanshasPC:~/Program-4$ npm init -y
Wrote to /home/1rv24mc012_akanksha/Program-4/package.json:
```

```
{
  "name": "program-4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Step 5: Install the dependency

npm install express

```
1rv24mc012_akanksha@akanshasPC:~/Program-4$ npm install express
```

```
added 68 packages, and audited 69 packages in 7s
```

```
18 packages are looking for funding  
  run `npm fund` for details
```

```
found 0 vulnerabilities
```

Step 6: Create a new repository on the github account

Profile → repositories → new → give a name → create

Step 7: Getting Personal Access Token (PAT)

Profile → Settings → Developer Settings → Personal Access Token → PAT(Classic) →
Generate a new token → Select scopes: repo (required) → generate → copy the token

Step 8: Initialize Git

git init

git add .

git commit -m "Initial commit"

Now Git is ready.

```
1rv24mc012_akanksha@akanshasPC:~/Program-4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1rv24mc012_akanksha/Program-4/.git/
1rv24mc012_akanksha@akanshasPC:~/Program-4$ git add .
1rv24mc012_akanksha@akanshasPC:~/Program-4$ git commit -m "Initial Commit"
[master (root-commit) 53bbcc1] Initial Commit
636 files changed, 72748 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
create mode 100644 node_modules/body-parser/package.json
create mode 100644 node_modules/bytes/History.md
create mode 100644 node_modules/bytes/LICENSE
create mode 100644 node_modules/bytes/Readme.md
create mode 100644 node_modules/bytes/index.js
create mode 100644 node_modules/bytes/package.json
create mode 100644 node_modules/call-bind-apply-helpers/.eslintrc
```

Step 9: Connect Local Folder to the GitHub Repo

Run these commands in the project folder

```
git remote add origin https://github.com/<username>/<repo-name>.git
```

```
git branch -M main
```

```
git push -u origin main
```

```
1rv24mc012_akanksha@akanshasPC:~/Program-4$ git remote add origin https://github.com/Akanksha2124/DockIntegration.git
1rv24mc012_akanksha@akanshasPC:~/Program-4$ git branch -M main
1rv24mc012_akanksha@akanshasPC:~/Program-4$ git push -u origin main
Username for 'https://github.com': Akanksha2124
Password for 'https://Akanksha2124@github.com':
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 12 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 726.18 KiB | 5.81 MiB/s, done.
Total 719 (delta 142), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (142/142), done.
To https://github.com/Akanksha2124/DockIntegration.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

The top screenshot shows a GitHub repository page for 'Akanksha2124 / DockIntegration'. The repository is public and has one branch, 'main'. The commit history shows an initial commit with files: node_modules, Dockerfile, index.js, package-lock.json, and package.json. The README section is empty, with a prompt to 'Add a README'. The right sidebar shows repository statistics: 0 stars, 0 watching, 0 forks, 0 releases, 0 packages, and a language chart showing JavaScript at 57.2% and Dockerfile at 42.8%.

The bottom screenshot shows a local browser window with the URL 'http://localhost:3000'. The page displays 'Heloo from git +Docker Integration'.